AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-10-21 | Entry window 21 October 2026 (full day) | 18 records screened in scope

18 RECORDS IN SCOPE (86 EXCLUDED)

9 OF 10 SUBTOPIC BANDS POPULATED

5 EFFECT ESTIMATES WITH A CI

3 TIER-A RECORDS

5 RECORDS CARRIED ON METADATA ALONE

Signal of the day

26 million Medicare beneficiaries: atrial fibrillation tracks sulfate and combustion $PM_{2.5}$, not biomass smoke or secondary organics.

Source-apportioned $PM_{2.5}$ (Multilinear Engine on speciation-network data, inverse-distance weighted to ZIP code) against 4.44 million incident AF cases over 171.6 million person-years, 2000–2019. Per IQR: secondary sulfate HR **1.056** (1.053–1.058), heavy oil/coal combustion **1.048**, secondary nitrate 1.032, vehicle 1.029; biomass burning and dust modest; SOA null nationally but positive in the West. **Where it is weak:** source fields are interpolated from sparse speciation monitors over 100 km, so source contrasts are largely regional contrasts, and per-IQR scaling differs by source — this is not a per-microgram toxicity ranking. Claims-based AF misses paroxysmal disease. **Why it matters here:** optical low-cost networks report mass. If composition carries the cardiac signal, a dense mass network needs a sparse speciation backbone to be epidemiologically useful.

What changed today

- **Particle size matters for spirometry.** China Pulmonary Health Study, 50,991 adults: per IQR PM_1 FEV_1 -36.8 **mL** (-45.9 to -27.8); large-airway decrements grow as size falls, while the coarse fraction $PM_{2.5–10}$ gives the largest small-airway ($FEF_{25–75}$) decrements.
- **An outcome-weighted air index.** 1.21 million AMI cases, 325 cities: a composite Air Health Index ($PM_{2.5}$, NO_2, O_3, temperature) gives +5.97 % per IQR vs +3.33 % for AQHI and +2.04 % for AQI — in-sample advantage expected, but externally validated.
- **Implausibly large cancer-prognosis HRs.** Cervical cancer, 3,758 women: $PM_{2.5}$ recurrence SHR 1.93, death HR 2.55 per IQR on 113 deaths — read as confounding by referral geography until replicated.
- **Magnetite inventory.** Respirable magnetite from Chinese coal power: 2,829 $t\,y^{-1}$; two provinces with 4 % of output could cut >31 % of emissions by tightening standards.
- **PubMed was nearly silent.** Only 2 records carried a 21 October entry date in PubMed; Europe PMC and the Crossref ISSN sweep carried the issue, and five ACS *Environment & Health* papers arrived via Europe PMC with PMIDs assigned the next day.

Window continuity

The 20 October issue closed with `last_entry_date` at 20 October. This issue collects **21 October in full**, continuous with it. The scheduled run for 21 October (04:0x UTC 22 October) did not fire; this run started 18:02 CDT on 22 October, before the 22:00 cut-off, so 22 October is *not* opened and is left to the next run at or after 22:00 CDT. The W43 weekly (18–24 October) is due Saturday 24 October.

Corpus analytics

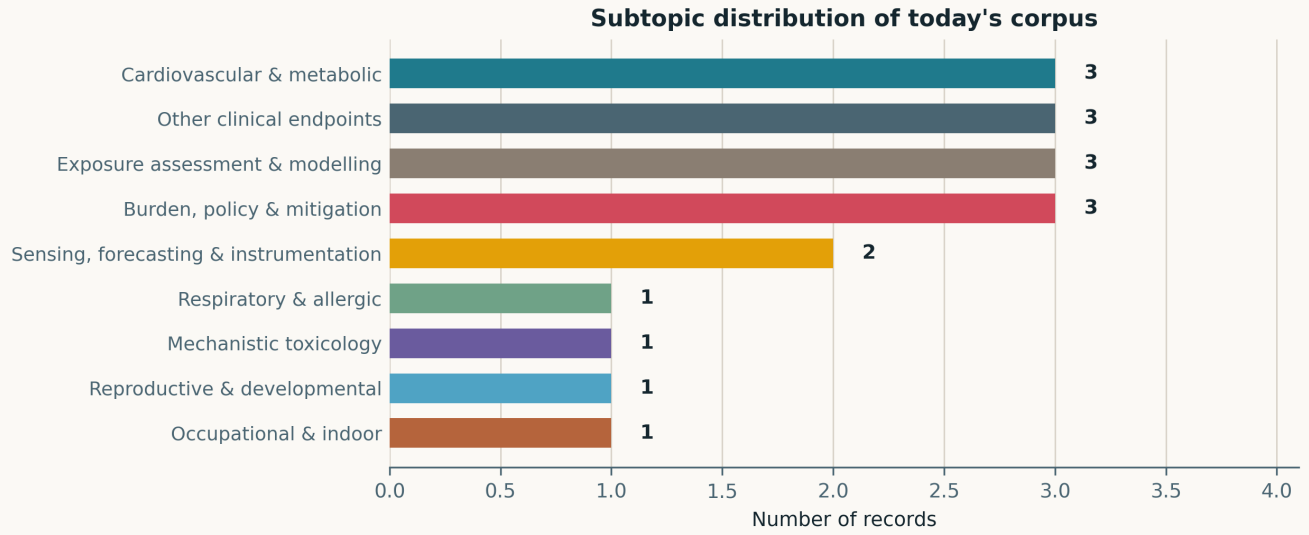


Fig. Subtopic assignment of the 18 in-scope records across nine bands. Cardiovascular, other clinical endpoints, exposure assessment and burden/mitigation hold three each; sensing two; neuro/mental health is the only empty band.

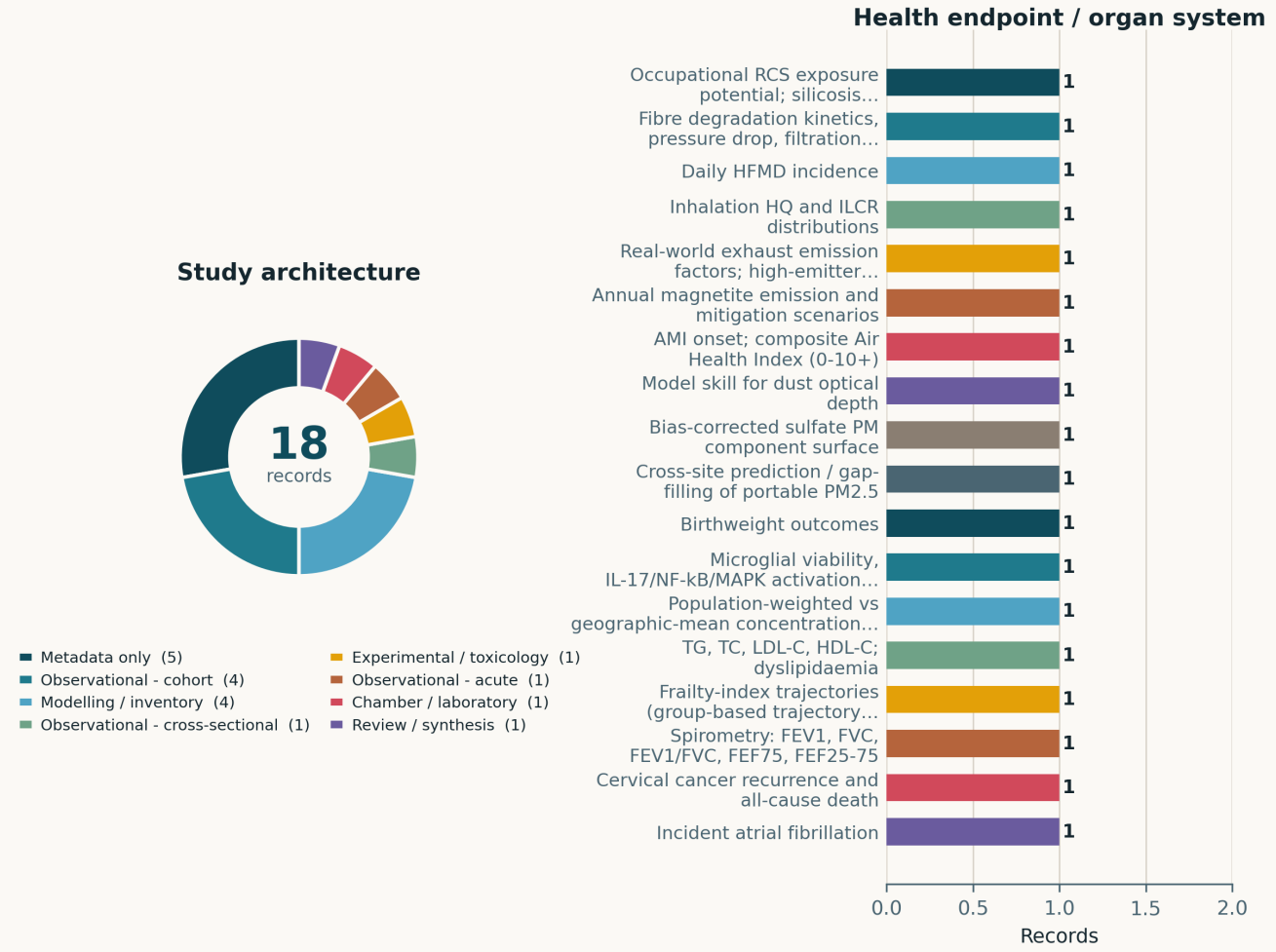


Fig. Left — architecture: four cohorts, four modelling/inventory records, one acute time-series, one cross-sectional, one toxicology, one laboratory, one review; **metadata-only 5 (28 %)**, day-0 Elsevier and one ACS record. Right — endpoints span AF, AMI, lipids, spirometry, frailty, cervical cancer prognosis, HFMD and birthweight.

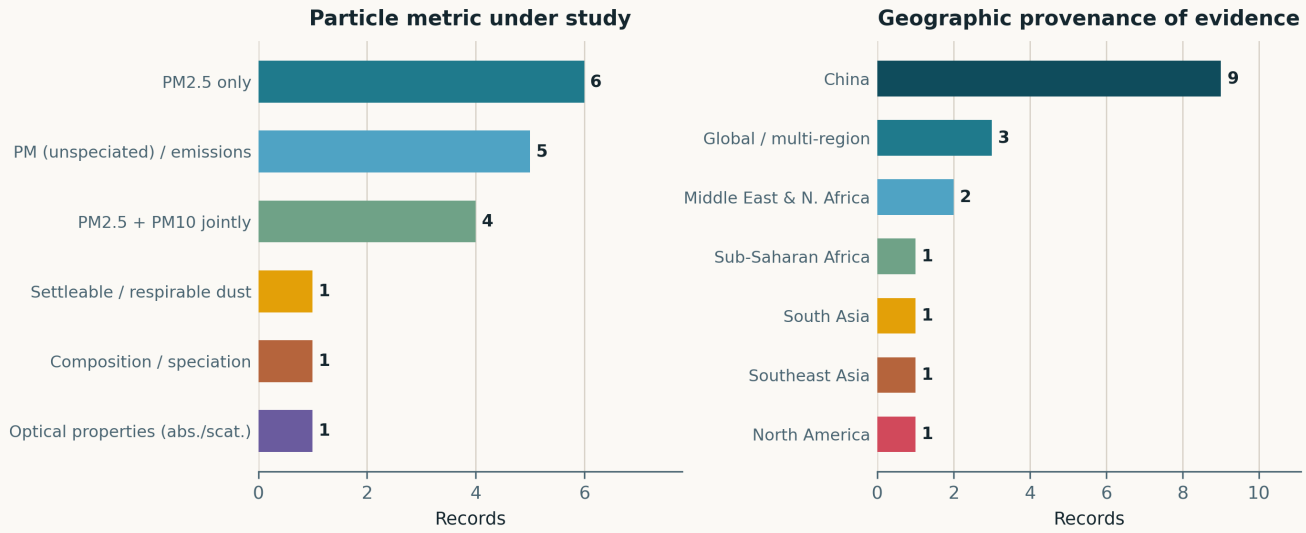


Fig. Left — particle metric: $PM_{2.5}$-only 6, joint $PM_{2.5}$+PM_{10} 4, unspeciated/emissions 5, one composition record (sulfate), one optical (dust optical depth), one respirable dust. Right — geography: China 9 dominates; MENA 2; one each for the US, Southeast Asia, South Asia, Nigeria.

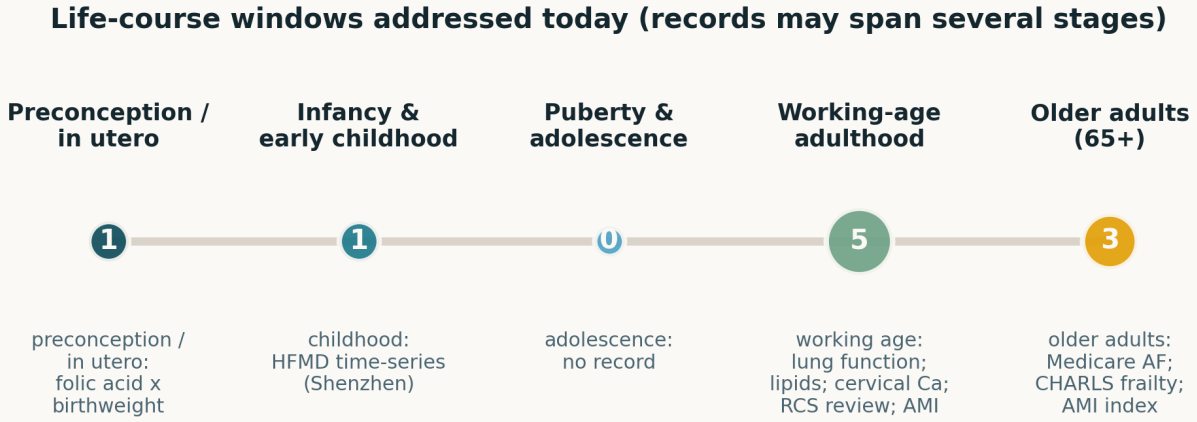


Fig. Life-course windows: working age leads (spirometry, lipids, cervical cancer, silica, AMI); three older-adult records (Medicare AF, CHARLS frailty, AMI); one in utero (folic acid, metadata only), one childhood (HFMD); no adolescent record.

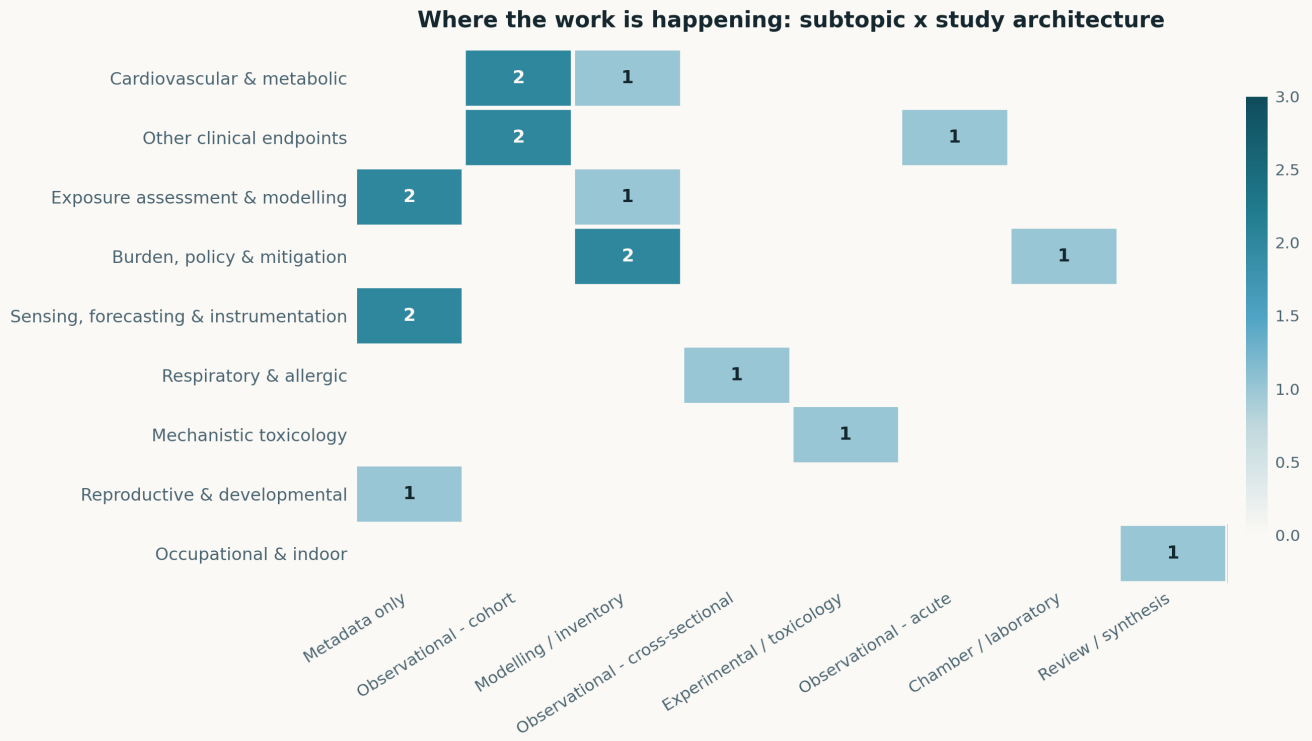


Fig. Subtopic × study architecture. Cardiovascular sits in cohorts and health-impact modelling; exposure assessment and sensing fall mostly in metadata-only cells today, which limits what the measurement side of this issue can assert.

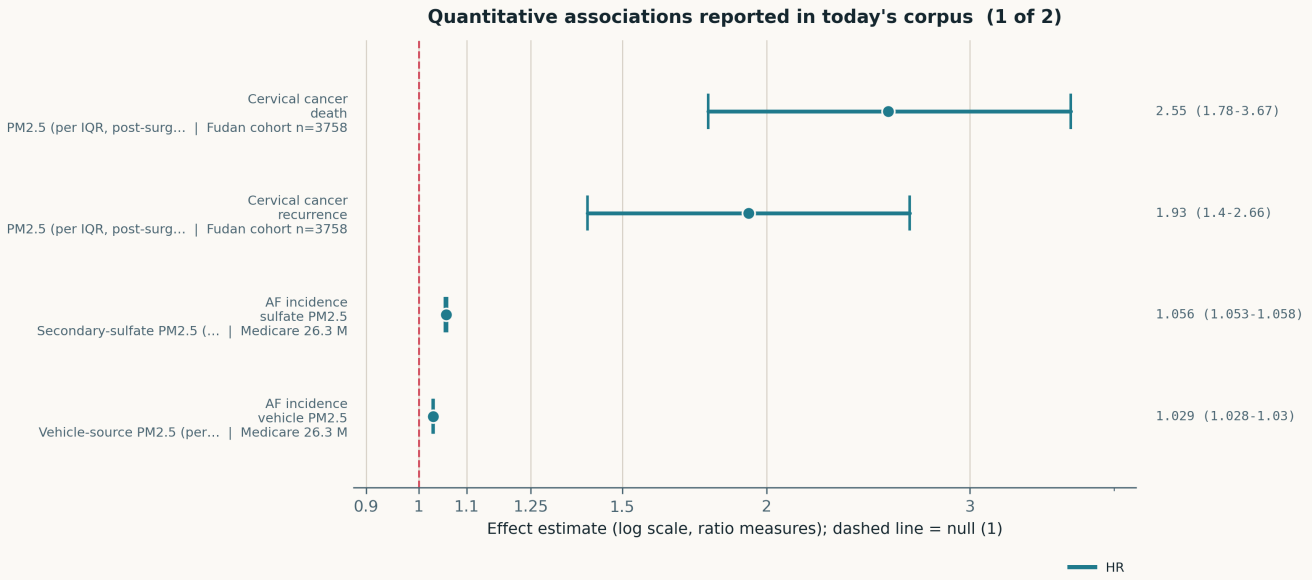


Fig. Ratio scale, four estimates from two studies: source-specific $PM_{2.5}$ (secondary sulfate, vehicle) and incident AF per IQR in Medicare; $PM_{2.5}$ per IQR and cervical cancer recurrence (Fine–Gray SHR) and death in a single-centre cohort. Per-IQR contrasts differ by source and study. Not poolable.

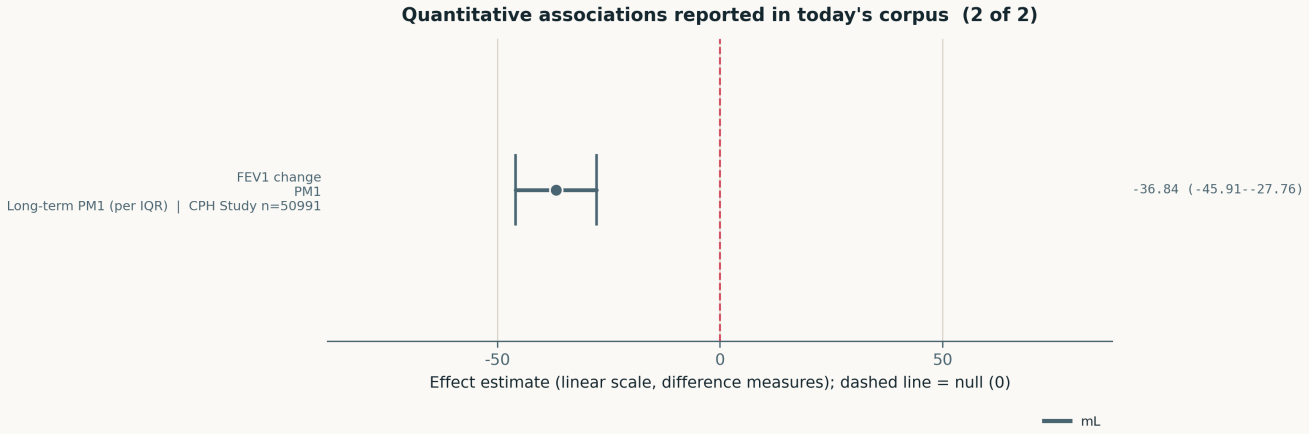


Fig. Difference scale: FEV_1 change (mL) per IQR long-term PM_1, China Pulmonary Health Study (n = 50,991). Single estimate shown for scale; the coarse-fraction small-airway estimates are not given with CIs in the abstract.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Cardiovascular & metabolic — 3 records

Zhu source-specific $PM_{2.5}$ atrial fibrillation Medicare cohort 2026 Environ Health (ACS)

Source-Specific $PM_{2.5}$ Exposure and Atrial Fibrillation Risk in U.S. Older Adults: A Nationwide Cohort Study

26.3 million Medicare beneficiaries, 2000-2019, 171.6 M person-years, 4.44 M incident AF. Per IQR of source-specific $PM_{2.5}$: secondary sulfate HR 1.056 (1.053-1.058), heavy oil/coal combustion 1.048 (1.046-1.050), secondary nitrate 1.032 (1.030-1.034), vehicle 1.029 (1.028-1.030); biomass burning and dust modest, SOA null overall but positive in the West. Limitation: source profiles come from sparse speciation monitors interpolated by inverse distance over 100 km, so source contrasts are largely regional contrasts; sulfate and nitrate fields are smooth and the per-IQR scaling differs by source, so 'sulfate is most toxic' is not a per-microgram statement. Claims-based AF under-ascertains paroxysmal AF. Relevance: high - speciation, not mass, is what optical LCS cannot give; source-resolved risk raises the value of co-located filter or ACSM nodes.

doi: 10.1021/envhealth.5c00638

Sun long-term air pollution lipid profiles Shanxi repeated measures 2026 Front Public Health

Long-term exposure to ambient air pollution and lipid profiles: a longitudinal study in a heavily industrialized region of northern China

Repeated-measures study of 9,040 adults with 19,113 lipid measurements from a health-screening programme, 2019-2023. Per 10 $\mu g\,m^{-3}$ $PM_{2.5}$: TG +1.85%, TC +1.70%, LDL-C +1.81%, HDL-C -1.55% (linear mixed models); PM_{10}, NO_2, SO_2 and O_3 in the same direction, CO weaker; age modified the HDL-C association after FDR correction. Limitation: a health-screening population self-selects; CHAP is itself trained on the sparse Shanxi monitor network that underrepresents coal-industry hotspots, and with five correlated pollutants the '$PM_{2.5}$ most pronounced' ranking is not identified from single-pollutant models. Lipids are sensitive to fasting status, not mentioned. Relevance: moderate - industrial-region exposure surfaces are where LCS densification would most change CHAP errors.

doi: 10.3389/fpubh.2026.1950282

Du Air Health Index acute myocardial infarction 325 cities 2026 Environ Sci Technol

Establishing an Air Health Index for Predicting the Risk of Acute Myocardial Infarction Onset Associated with Air Pollution and Ambient Temperature

1,207,625 AMI cases across 325 Chinese cities; excess risks from $PM_{2.5}$, NO_2, O_3 and temperature aggregated into a normalised AHI. Per IQR, AHI associated with +5.97% (5.32-6.63%) AMI incidence vs +3.33% (2.93-3.72%) for AQHI and +2.04% (1.73-2.35%) for AQI; near-linear, no threshold; validated on an independent external dataset; 8,534 per 100,000 AMI cases attributed. Limitation: the index is fitted to the same outcome it is scored on, so in-sample superiority over AQI/AQHI is expected; IQR-scaled comparisons across indices with different distributions are not like-for-like, and 'attributable' fractions for a composite that includes temperature mix avoidable and non-avoidable risk. Relevance: high - an outcome-weighted index is a natural output layer for real-time LCS networks, provided the $PM_{2.5}$ input is calibrated.

doi: 10.1021/acs.est.6c07635

Respiratory & allergic — 1 record

Li size-specific PM lung function China Pulmonary Health Study 2026 Environ Health (ACS)

Associations between Size-Specific Particulate Matter and Lung Function: Results from the China Pulmonary Health Study

50,991 adults >=20 y from the China Pulmonary Health Study; multivariable linear models on long-term size-resolved PM. Per IQR PM_1: FEV1 -36.84 mL (-45.91 to -27.76), FVC -34.63 mL (-45.48 to -23.79); large-airway decrements grew as particle size fell, whereas the coarse fraction $PM_{2.5}$-10 gave the largest small-airway (FEF75, FEF25-75) decrements. Limitation: size fractions are derived from correlated modelled surfaces (PM_1 and $PM_{2.5}$ share most of their mass), and the abstract reports no multi-fraction model, so 'size-specific' ranking may reflect differential exposure-model error rather than biology; cross-sectional. Relevance: high - optical LCS report PM_1 and coarse bins natively, and this is evidence that those channels carry distinct health signal worth calibrating, not discarding.

doi: 10.1021/envhealth.5c00676

Reproductive & developmental — 1 record

Ge preconception folic acid air pollution birthweight 2026 Environ Health (ACS)

Preconception Folic Acid Supplementation Associated with Attenuated Effects of Air Pollution on Birthweight Outcomes

METADATA ONLY. Title and authors only (Ge, Zhang, Liao ... Berhane, Bai et al.); no abstract retrievable from PubMed, Europe PMC, Crossref or Semantic Scholar at harvest. The title claims effect modification of the air pollution-birthweight association by preconception folic acid use. Pollutants, cohort, n and effect sizes cannot be stated. Limitation: unverifiable beyond the title; supplement use is self-reported and strongly socially patterned, so 'attenuation' by folate is exposed to confounding by SES and antenatal-care quality unless handled explicitly. Relevance: low - no sensing content evident.

doi: 10.1021/envhealth.5c00779

Mechanistic toxicology — 1 record

Liu $PM_{2.5}$ IL-17 microglia neuroinflammation 2026 Environ Health (ACS)

IL-17 Signaling Pathway: A Potential Signaling Pathway Mediating Neuroinflammation Induced by $PM_{2.5}$

BV2 microglia exposed to $PM_{2.5}$ lost viability and raised inflammatory cytokines; microarray DEGs enriched for inflammatory GO terms and the KEGG IL-17 pathway with downstream NF-kB and MAPK. An anti-IL-17A antibody reversed cytokine release, microglial activation and behavioural deficits (implying an in vivo arm). Limitation: the abstract gives no $PM_{2.5}$ source, composition or dose, so environmental relevance cannot be judged; BV2 is an immortalised murine line with an exaggerated inflammatory phenotype, and IL-17 is usually T-cell/gamma-delta derived, so microglial IL-17 signalling needs cell-type attribution. Relevance: low - mechanistic only.

doi: 10.1021/envhealth.6c00080

Exposure assessment & modelling — 3 records

Mai population-weighted $PM_{2.5}$ O_3 China 2015-2024 preprint 2026 Research Square (preprint)

Population Weighted Ambient $PM_{2.5}$ and O_3 Concentrations in China from 2015 to 2024

Ecological geospatial analysis of annual ~1 km CHAP $PM_{2.5}/O_3$ rasters matched to WorldPop counts, 2015-2024. Population-weighted $PM_{2.5}$ fell from 53.06 to 31.25 $\mu g\,m^{-3}$ (-41.1%) while population-weighted O_3 rose from 77.06 to 96.75 $\mu g\,m^{-3}$ (+25.6%); the population-weighted minus geographic gap narrowed for $PM_{2.5}$ (5.77 to 2.43) and widened for O_3 (1.01 to 2.72). 97.4% of 0.2-degree blocks show falling $PM_{2.5}$ with rising O_3 at a 0.5 $\mu g\,m^{-3}$/y threshold. Limitation: purely descriptive and inherits CHAP's ML smoothing; no uncertainty on the weighted means; preprint, not peer reviewed. Relevance: moderate - the narrowing urban-rural $PM_{2.5}$ gap shifts LCS value toward peri-urban and rural siting.

doi: 10.21203/rs.3.rs-10964316/v1

Tripathi satellite-reanalysis sulfate aerosol India ground measurements 2026 Environ Pollut

Improving Satellite-Reanalysis Sulfate Aerosol Estimates over India Using Ground Measurements

METADATA ONLY. Title and authors only (Tripathi, Sahu, Sharma, Gupta); no abstract indexed at harvest. The title indicates ground measurements used to correct satellite-reanalysis (likely MERRA-2/CAMS-type) sulfate aerosol over India. Data sources, method and skill cannot be stated. Limitation: unverifiable beyond the title; Indian speciation networks are sparse and urban-biased, so any correction trained on them risks transferring urban bias to rural cells. Relevance: moderate - component-resolved surfaces are the complement LCS mass networks lack; pairs with the sulfate-dominant AF signal in today's Medicare cohort.

doi: 10.1016/j.envpol.2026.129391

Dayanandana CMIP6 dust optical depth MENA evaluation 2026 Atmos Environ

Evaluation of Dust Optical Depth in CMIP6 Simulations over the Middle East-North Africa Region

METADATA ONLY. Title and authors only (Dayanandana, Parottil, Abbas, Al-Harrasi et al.); no abstract at harvest. Evaluates CMIP6 dust optical depth over MENA, presumably against MODIS/MERRA-2 DOD. Models, metrics and biases cannot be stated. Limitation: unverifiable beyond the title; column optical depth maps to surface PM_{10} only through assumed vertical profiles and size distributions, so climate-model DOD skill says little about surface exposure. Relevance: low-moderate - MENA dust is where optical LCS mass error from coarse particles is largest.

doi: 10.1016/j.atmosenv.2026.122431

Sensing, forecasting & instrumentation — 2 records

Pyae Tin Saw portable $PM_{2.5}$ cross-site prediction fixed station 2026 Environ Pollut

Short-term $PM_{2.5}$ data continuity: Cross-site prediction of portable measurements using a nearby fixed monitoring station

METADATA ONLY. Title and authors only (Pyae Tin Saw, Kallawicha); day-0 Elsevier deposit with no abstract anywhere at harvest. The title describes predicting short-term portable $PM_{2.5}$ records from a nearby fixed monitoring station to maintain data continuity. Sensor model, distance, n, and skill metrics cannot be stated. Limitation: unverifiable beyond the title; cross-site transfer functions are typically valid only within the regional-background regime and fail for local sources, which is exactly when portable data matter. Relevance: high - gap-filling an LCS or portable record against a regulatory anchor is a core network-operations problem.

doi: 10.1016/j.envpol.2026.129384

Ghaffarpasand Pope vehicle remote sensing real-world emissions high emitters 2026 Atmos Pollut Res

Emerging Evidence on Real-World Exhaust Emissions and High Emitting Vehicles from Remote Sensing Observations

METADATA ONLY. Title and authors only (Ghaffarpasand, Pope); no abstract at harvest. Appears to synthesise vehicle emission remote-sensing (VERSS) evidence on real-world exhaust and high emitters. Whether PM/smoke opacity or only NOx/CO/HC is covered cannot be stated. Limitation: unverifiable beyond the title; VERSS PM is inferred from UV opacity with large per-pass uncertainty, and single-pass snapshots overweight hot-running conditions. Relevance: moderate - high-emitter fractions explain the kerbside spikes roadside LCS record but cannot attribute.

doi: 10.1016/j.apr.2026.103244

Occupational & indoor — 1 record

Alquwayi respirable crystalline silica occupational exposure Saudi Arabia review 2026 Front Public Health

Potential risk of occupational exposure to respirable crystalline silica in Saudi Arabia: a comprehensive review of the current status and future challenges

Single-author review integrating literature, labour and mining statistics, mineralogy and regulation through a 'silica hazard triangle' (workforce, materials, activities). Concludes that mining, construction, cement, glass, engineered stone and photovoltaic manufacturing expansion plus arid-climate dust raise RCS exposure potential, while measured exposure, epidemiology and surveillance data are scarce. Limitation: no new measurements and no quantitative exposure estimates - the conclusion is an inference from sector size, not from data. Relevance: low-moderate - optical LCS cannot speciate silica, so RCS monitoring needs gravimetric/FTIR sampling regardless.

doi: 10.3389/fpubh.2026.1918362

Burden, policy & mitigation — 3 records

Zhang respirable magnetite emissions coal-fired power China 2026 Environ Sci Technol

Emissions of Respirable Magnetite from the Coal-Fired Power Sector in China: Spatial Inequalities and Mitigation Pathways

Multifactor emission model for respirable magnetite from Chinese coal-fired power: 2,828.7 (2,307.1-3,371.4) t/y, spatially concentrated by tiered emission standards. Guizhou and Sichuan supply 4% of coal power but tightening their standards alone would cut >31% of national magnetite emissions. Limitation: emission factors for magnetite specifically (as opposed to total PM or Fe) rest on few stack measurements, and the uncertainty interval looks narrow for a species-level inventory; health relevance is asserted from brain-magnetite literature rather than exposure-response data. Relevance: moderate - magnetite is invisible to optical LCS mass; magnetic or filter-based speciation would be needed to verify hotspots.

doi: 10.1021/acs.est.6c08602

Idowu medical waste incinerator airborne heavy metals Monte Carlo Nigeria 2026 Research Square (preprint)

Monte Carlo-Based Probabilistic Health Risk Assessment of Airborne Heavy Metals from a Medical Waste Incinerator in Southwestern Nigeria

Particulate matter collected on passive glass slides around a medical-waste incinerator for two months; ten metals analysed, converted to airborne concentrations, and propagated through 10,000 Monte Carlo draws. Median HQ ~400, 95th percentile 850; exceedance probability near 100%; ILCR above benchmarks. Limitation: passive deposition slides measure settled mass, not airborne concentration - the conversion to C_air requires an assumed deposition velocity that dominates the result, and an HQ of 400 almost certainly reflects that conversion rather than real inhalation dose. Preprint. Relevance: moderate as a cautionary example - fence-line LCS or low-volume samplers would have given a defensible air concentration.

doi: 10.21203/rs.3.rs-11381994/v1

Jiang PLA melt-blown nonwoven filtration durability humidity airflow 2026 Langmuir

Service Durability of PLA Melt-Blown Nonwovens: Roles of Humidity and Airflow Velocity in Degradation and Filtration Performance

Full-factorial service-condition study (humidity, airflow velocity, PM loading, natural microbes) on PLA melt-blown filter media. Degradation stays early-stage, first-order, k up to 3.88e-2 /h, surface-erosion dominated. Humidity drives degradation (90% of short-term water-contact-angle variance; 77% of yield-strength and 91% of pressure-drop variance long-term); airflow velocity governs dynamic filtration via particle-media contact. Limitation: the abstract gives no filtration efficiency numbers by particle size, so it is unclear whether MPPS performance falls within a realistic mask or HVAC service life. Relevance: low - mitigation materials; humid-climate filter life matters for purifier intervention trials.

doi: 10.1021/acs.langmuir.6c03984

Other clinical endpoints — 3 records

Lin air pollution cervical cancer recurrence mortality Fudan cohort 2026 Environ Health (ACS)

Long-Term Exposure to Ambient Air Pollution and Cervical Cancer Outcomes in Chinese Women: A Prospective Study on Recurrence and Mortality Risks

3,758 women with primary cervical cancer (2019-2022), median follow-up ~3 y; 217 recurrences, 113 deaths. Per IQR $PM_{2.5}$: recurrence SHR 1.93 (1.40-2.66, Fine-Gray), death HR 2.55 (1.78-3.67); PM_{10} SHR 1.65 (1.13-2.42) and HR 2.21 (1.38-3.54); NO_2 and SO_2 also positive, O_3 null. Limitation: hazard ratios near 2-2.5 per IQR from 113 deaths are implausibly large for a residential long-term contrast and point to confounding by referral geography - patients from outside Shanghai (higher PM) likely differ in stage, HPV management and follow-up intensity; single-centre, short follow-up, no adjustment for distance to the hospital reported in the abstract. Relevance: low - model-based residential exposure, no sensing implication.

doi: 10.1021/envhealth.6c00011

Lin $PM_{2.5}$ frailty trajectories CHARLS 2026 J Occup Environ Med

The association of particulate matter 2.5 exposure with frailty trajectories in middle-aged and older Chinese adults and the potential modification by lifestyle and socioeconomic status

10,659 CHARLS participants; deficit-accumulation frailty index, three GBTM trajectories (stable low 64%, increasing to frailty 29%, continuously worsening 7%). Per 10 $\mu g\,m^{-3}$ $PM_{2.5}$, odds of the two adverse trajectories rose 15.2% and 12.3% (ordered logistic; no CIs in the abstract). Higher SES attenuated the association (RERI -0.067); unhealthy sleep duration amplified it (RERI 0.163). Limitation: GBTM classes are model artefacts with posterior misclassification carried into a second-stage regression without uncertainty propagation; CHARLS addresses are coarse (city/county), so exposure contrast is mostly between regions with different healthcare and diet. Relevance: low - regional satellite exposure, no sensing implication.

doi: 10.1097/jom.0000000000003913

Jiang air pollution meteorology hand foot mouth disease Shenzhen DLNM 2026 Front Public Health

Ambient air pollution and meteorological factors associated with hand, foot, and mouth disease risk in Shenzhen, China: a time-series study

Shenzhen daily HFMD counts 2013-2022 with six pollutants and meteorology in DLNMs, including pre/post EV71 vaccine and COVID-19 periods. $PM_{2.5}$ and PM_{10} showed risk peaks at low concentrations and 'protective' effects at extreme highs; SO_2 and CO raised cumulative risk at extremes; temperature showed a dual-peak pattern and high humidity raised risk. No effect sizes with CIs in the abstract. Limitation: HFMD is a contact-transmitted enteroviral disease with strong seasonality driven by school terms; non-monotone, sign-flipping PM curves are the signature of residual seasonal confounding and DLNM over-flexibility, not biology. Relevance: low.

doi: 10.3389/fpubh.2026.1920424

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Du Air Health Index acute myocardial infarction 325 cities	Establishing an Air Health Index for Predicting the Risk of Acute Myocardial Infarction Onset Associated with Air Pollution and Ambient Temperature	*Environ Sci Technol*	Health impact model	Daily $PM_{2.5}$ (with NO2, O3, temperature) from high-resolution surfaces	China (325 cities)
Sun long-term air pollution lipid profiles Shanxi repeated measures	Long-term exposure to ambient air pollution and lipid profiles: a longitudinal study in a heavily industrialized region of northern China	*Front Public Health*	Prospective cohort	3-y mean $PM_{2.5}$, PM_{10} (ChinaHighAirPollutants, geocoded residence)	China (Shanxi Province)
Zhu source-specific $PM_{2.5}$ atrial fibrillation Medicare cohort	Source-Specific $PM_{2.5}$ Exposure and Atrial Fibrillation Risk in U.S. Older Adults: A Nationwide Cohort Study	*Environ Health (ACS)*	Retrospective cohort	Source-apportioned $PM_{2.5}$ (Multilinear Engine on CSN/IMPROVE speciation; IDW to ZIP code, 100 km)	USA (nationwide Medicare)
Reproductive & developmental					
Ge preconception folic acid air pollution birthweight	Preconception Folic Acid Supplementation Associated with Attenuated Effects of Air Pollution on Birthweight Outcomes	*Environ Health (ACS)*	Metadata only (no abstract)	Ambient air pollution (metrics not stated in record)	China (inferred from author group; not stated in record)
Respiratory & allergic					
Li size-specific PM lung function China Pulmonary Health Study	Associations between Size-Specific Particulate Matter and Lung Function: Results from the China Pulmonary Health Study	*Environ Health (ACS)*	Observational - cross-sectional	Long-term PM1, PM1-2.5, $PM_{2.5}$, $PM_{2.5}$-10, PM_{10} (per IQR)	China (national, CPH Study)
Mechanistic toxicology					
Liu $PM_{2.5}$ IL-17 microglia neuroinflammation	IL-17 Signaling Pathway: A Potential Signaling Pathway Mediating Neuroinflammation Induced by $PM_{2.5}$	*Environ Health (ACS)*	Animal + in vitro	Collected $PM_{2.5}$ suspension (dose not stated in abstract)	Laboratory
Occupational & indoor					
Alquwayi respirable crystalline silica occupational exposure Saudi Arabia review	Potential risk of occupational exposure to respirable crystalline silica in Saudi Arabia: a comprehensive review of the current status and future challenges	*Front Public Health*	Narrative review	Respirable crystalline silica (respirable dust fraction)	Saudi Arabia
Sensing, forecasting & instrumentation					
Ghaffarpasand Pope vehicle remote sensing real-world emissions high emitters	Emerging Evidence on Real-World Exhaust Emissions and High Emitting Vehicles from Remote Sensing Observations	*Atmos Pollut Res*	Metadata only (no abstract)	On-road remote sensing of exhaust (pollutants not stated in record)	Not stated (review)
Pyae Tin Saw portable $PM_{2.5}$ cross-site prediction fixed station	Short-term $PM_{2.5}$ data continuity: Cross-site prediction of portable measurements using a nearby fixed monitoring station	*Environ Pollut*	Metadata only (no abstract)	Portable $PM_{2.5}$ monitor vs nearby fixed-station $PM_{2.5}$	Thailand (inferred from author affiliation; not stated in record)
Exposure assessment & modelling					
Dayanandana CMIP6 dust optical depth MENA evaluation	Evaluation of Dust Optical Depth in CMIP6 Simulations over the Middle East-North Africa Region	*Atmos Environ*	Metadata only (no abstract)	Dust optical depth (CMIP6 models vs observations)	Multi-country (Middle East and North Africa)
Mai population-weighted $PM_{2.5}$ O3 China 2015-2024 preprint	Population Weighted Ambient $PM_{2.5}$ and O3 Concentrations in China from 2015 to 2024	*Research Square (preprint)*	Exposure model (secondary data)	Annual 1 km CHAP $PM_{2.5}$ and O3 weighted by WorldPop	China (national)
Tripathi satellite-reanalysis sulfate aerosol India ground measurements	Improving Satellite-Reanalysis Sulfate Aerosol Estimates over India Using Ground Measurements	*Environ Pollut*	Metadata only (no abstract)	Sulfate aerosol (satellite/reanalysis vs ground)	India
Burden, policy & mitigation					
Idowu medical waste incinerator airborne heavy metals Monte Carlo Nigeria	Monte Carlo-Based Probabilistic Health Risk Assessment of Airborne Heavy Metals from a Medical Waste Incinerator in Southwestern Nigeria	*Research Square (preprint)*	Health impact model	PM-bound heavy metals (passive glass-slide deposition samplers)	Nigeria (Ido-Ekiti)

First author	Focus	Journal	Design	Metric	Region
Jiang PLA melt-blown nonwoven filtration durability humidity airflow	Service Durability of PLA Melt-Blown Nonwovens: Roles of Humidity and Airflow Velocity in Degradation and Filtration Performance	*Langmuir*	Chamber / laboratory	Filtration of particulate matter by PLA melt-blown media	Laboratory
Zhang respirable magnetite emissions coal-fired power China	Emissions of Respirable Magnetite from the Coal-Fired Power Sector in China: Spatial Inequalities and Mitigation Pathways	*Environ Sci Technol*	Emission inventory	Respirable magnetite (Fe3O4) particle emissions	China (national, provincial)
Other clinical endpoints					
Jiang air pollution meteorology hand foot mouth disease Shenzhen DLNM	Ambient air pollution and meteorological factors associated with hand, foot, and mouth disease risk in Shenzhen, China: a time-series study	*Front Public Health*	Time-series	Daily city-wide $PM_{2.5}$ and PM_{10} (fixed monitors)	China (Shenzhen)
Lin $PM_{2.5}$ frailty trajectories CHARLS	The association of particulate matter 2.5 exposure with frailty trajectories in middle-aged and older Chinese adults and the potential modification by lifestyle and socioeconomic status	*J Occup Environ Med*	Prospective cohort	Satellite-derived $PM_{2.5}$ (Dalhousie/ACAG surface, per 10 $\mu g/m^3$)	China (national, CHARLS)
Lin air pollution cervical cancer recurrence mortality Fudan cohort	Long-Term Exposure to Ambient Air Pollution and Cervical Cancer Outcomes in Chinese Women: A Prospective Study on Recurrence and Mortality Risks	*Environ Health (ACS)*	Prospective cohort	Residential $PM_{2.5}$ and PM_{10} from high-resolution spatiotemporal models (post-surgery)	China (Shanghai, single hospital)

Provenance and method

Entry window **21 October 2026**, single day (Wednesday). Harvester legs: **PubMed** health **2**, sensing **0** (connector EDAT query confirms 2 total — indexing lull); **Europe PMC 53** (2 AGRICOLA dropped); **Crossref by ISSN 61**; **arXiv** HTTP 429; **OpenAlex** HTTP 429. 113 unique, 10 already carried, **103 fresh**; PubMed connector (EDAT 21 Oct) 2 PMIDs, 0 appended. Europe PMC records lacking DOI were resolved by Crossref title match (five returned supplementary-file `.s001` DOIs, stripped and verified) and their abstracts recovered from PubMed by DOI. **Consensus** calibration sweep: 10 hits, 1 candidate (Ginsburg 2026, PurpleAir global calibration) — **rejected by the DOI gate as already shipped 29 July**. **104** candidates screened, **18 admitted**, **86 rejected** with logged reasons. **DOI gate** 0 fail / 0 warn after the duplicate was removed. **Trial watch** 21–22 October: 2 registrations updated (AIRWEIGHS purifier RCT NCT02763917, primary completion passed; Taif COPD survey). **5 carried on metadata alone**. **Scholar Gateway** not called. Abstracts remain the copyright of their publishers.